Forensic Science

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Forensic Science (A08022)

Short Title: Forensic Science
Faculty Science and Computing
Department Science
Cluster Analytical Science
Author KGRENNAN
Credits 5 **Level:** Introductory

Description of Module / Aims

This module will introduce the student to the world of the forensic scientist. The scientific principles associated with a range of forensic disciplines in the areas of impression, trace, biological and chemical evidence will be introduced. Students will learn how to employ simple tests to identify suspect substances in crimes related to the above disciplines. Students will learn how to interpret their findings and apply their conclusions methodically to a variety of criminal scenarios. Case studies detailing how these forensic techniques helped to solve true crimes will reinforce the scientific principles involved. The procedures involved in the collection of evidence at a crime scene will also be demonstrated, as will the procedures needed to maintain the chain of custody of evidence.

Programmes

FORS-0017	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 2 / E
FORS-0017	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	1 / 2 / E
FORS-0017	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	1 / 2 / E
FORS-0017	BSc in Pharmaceutical Science (WD_SPHSC_D)	1 / 2 / E

Indicative Content

- Fingerprints: historical development, collection, interpreting fingerprint patterns, databases
- Impressions: collecting and interpreting footprint, tyre mark and bite mark patterns
- Anthropology: establishing the biological profile of a victim through the analysis of bones & teeth
- Questioned documents: handwriting analysis & methods to detect document alterations
- Toxicology: mode of action of poisons & drugs of abuse, presumptive & confirmatory tests for drugs
- Serology: presumptive and confirmatory tests for body fluids, and ABO blood grouping
- Bloodstain pattern analysis: establishing the order of events at the crime scene
- DNA profiling: DNA structure & function, DNA profiling protocols, databases, random match probability
- Trace evidence: collecting, analysing and interpreting hair, fibre, glass, soil and paint evidence
- Crime scene investigation: crime scene procedures, chain of custody of evidence, preventing evidence contamination

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify methods to collect a variety of physical evidence from a crime scene following the rules of evidence.
2. Explain the practices involved in the routine testing of impression, trace, biological and chemical evidence.
3. Employ simple forensic techniques to test on impression, trace, biological and chemical evidence.
4. Interpret the results of a range of forensic tests on impression, trace, biological and chemical evidence.
5. Discuss the outcomes of case studies where forensic science played a key role in the criminal investigation process.

Learning and Teaching Methods

- Lectures.
- Tutorials.
- In-class demonstrations.
- Hands-on practical activities.
- Additional e-learning resources provided via Moodle.

Assessment Methods

	Weighing	**Outcomes Assessed**
Continuous Assessment	100%	
Tutorial/Problem Sheets	70%	1,2,3,4
Case Studies	30%	5

Assessment Criteria

<40%: The learner will demonstrate little or no understanding of the fundamentals of the learning outcomes. Will display a lack of competence with regards to an understanding of the basic practices of forensic science.

40%-49%: The learner will demonstrate limited understanding of the fundamentals of the learning outcomes. Will display some ability with regard to an understanding of the basic practices of forensic science.

50%-59%: As well as the above, the learner will demonstrate reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regard to an understanding of the basic practices of forensic science.

60%-69%: In addition, the learner will demonstrate more detailed knowledge of the learning outcomes. Will display good competence with regards to an understanding of the basic practices of forensic science.

70%-100%: All of the above to excellent levels. In addition, the learner will demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to an understanding of the basic practices of forensic science.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	12	
Independent Learning	99	

Essential Material

- "Forensic Science module Moodle page." http://moodle.wit.ie
- Jackson, A.R.W. _Forensic Science_. Harlow: Pearson Prentice Hall, 2008.

Supplementary Material

- Fraser, J.C. _Forensic Science: A Very Short Introduction_. Oxford: Oxford University Press, 2010.
- Langford, A.M., J. Dean, D. Holmes, A. Jones, R. Reed and Weyers, J. _Practical Skills in Forensic Science_. Harlow: Prentice Hall, 2010.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Computer Lab